



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	CSC3002
Course title (English)	C/C++ Programming
Course title (Chinese)	C/C++程序设计
Units	3
Language of Instruction	English
Description (English)	Building on the background from the first programming course "Introduction to Computer Science: Programming Methodology", we will move from the Python programming language to C++, and cover more advanced programming techniques such as recursion and dynamic memory management, explore advanced data structures and algorithms, compare procedural, functional, and object-oriented programming paradigms, and apply these tools to solving complex problems.
Description (Chinese)	本课程以C++为教学语言教授高阶的编程方法，除了必要的数据结构和算法，涵盖递归、对象、模板、流、指针、动态内存管理等高阶内容，并以C++为例对比过程型、函数型、以及面向对象型等不同类型的编程范式。

2. Prerequisites / Co-requisites

A. Prerequisites

CSC1001 Introduction to CS CSC1003 Intro to CS and JP CSC1005 Introduction to Computer Engin

B. Co-requisites

NA

3. Learning Outcomes



Upon completing this course, students will have acquired:

- Fundamental knowledge in computer science and engineering, including computer system and architecture, computer programming and software, etc., and the prerequisites for further computer science and engineering courses (K);
 - Knowledge of the C++ programming language, compilers and IDEs, etc., and programming skills, including designing, implementing, and testing various C++ programs (K&S);
 - Ability to identify, formulate, and solve practical problems using various computer programming techniques (S&V);
 - Understanding of the requirements of professional ethics (V)
- K: Knowledge outcomes; S: Skills outcomes; V: Values outcomes

4. Course syllabus

- Overview of the C++ programming language, compilers and IDEs
- Programming basics: functions, libraries, strings, streams, collections
- Object-oriented programming: objects, classes, inheritance
- The C++ memory model: pointers, arrays, dynamic memory management
- Recursion
- Algorithmic complexity analysis, efficiency and representation
- Linear structures: stacks, queues, vectors
- Advanced structures: maps, trees, sets, graphs
- Programming paradigms: procedural, functional, object-oriented, etc.

5. Assessment Scheme

Component/ method	% weight
Programming Project	50
Final Exam (Written exam)	50

6. Grade descriptor

General

Grade	Description
A	Demonstrates a deep understanding of the knowledge in computer science and



	engineering, especially in computer programming, in the written examination; demonstrates strong skills of designing, implementing, and testing various C++ programs, and great ability to identify, formulate, and solve practical problems using various computer programming techniques, with nearly no errors, in the programming assignments.
B	Demonstrates a general understanding of the knowledge in computer science and engineering, especially in computer programming, in the written examination; demonstrates necessary skills of designing, implementing, and testing various C++ programs, and average ability to identify, formulate, and solve practical problems using various computer programming techniques, with only minor errors, in the programming assignments.
C	Demonstrates a limited understanding of the knowledge in computer science and engineering, especially in computer programming, in the written examination; demonstrates basic skills of designing, implementing, and testing various C++ programs, and limited ability to identify, formulate, and solve practical problems using various computer programming techniques, with major errors sometimes, in the programming assignments.
D	Demonstrates a minimal understanding of the knowledge in computer science and engineering, especially in computer programming, in the written examination; demonstrates little skills of designing, implementing, and testing various C++ programs, and low ability to identify, formulate, and solve practical problems using various computer programming techniques, which often fail to perform as expected, in the programming assignments.
F	Demonstrates a poor understanding of the knowledge in computer science and engineering, especially in computer programming, in the written examination; fails to finish most of the programming assignments.

7. Feedback for evaluation

- Formal Course and Teaching Evaluation by students after finishing all the lectures;
- Informal feedback to the instructor and/or teaching assistant(s) through face-to-face discussions, video conferences, emails, etc.;
- Internal and/or external programme and/or course reviews of the school and/or the university.

8. Reading

9. Course components



Activity	Hours/week
Lecture	3
Readings & Project	6
Tutorial	1

10. Indicative teaching plan

Week	Content/ topic/ activity
001	Course introduction and overview of C++
002	Programming basics I: functions and libraries
003	Programming basics II: strings and streams
004	Programming basics III: collections
005	Object-oriented programming I: objects and classes
006	Pointers and arrays
007	Dynamic memory management
008	Recursion
009	Algorithm efficiency and representation
010	Linear structures: stacks, queues, vectors
011	Advanced structures I: maps, trees, sets
012	Advanced structures II: graphs
013	Object-oriented programming II: inheritance
014	Programming paradigms

11. Any other information

Declaration on the use of AI tools.

Free use of AI tools with no acknowledgment: This course requires frequent use of AI tools, and students can use these tools throughout the learning process without the need to explicitly cite or acknowledge them. The instructor may specify which AI tools are allowed and set proper regulations if needed. Students who do not follow the regulations will be considered academic dishonesty.

12. Version date

Version number	6
As of (date)	2024/11/18